

Vinay Tiwary

Embedded Firmware Engineer

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SUMMARY

Embedded Firmware Engineer with **3.5 years of experience** developing commercial HVAC controllers, smart appliances, IoT telemetry devices, and power-control systems. Skilled in embedded C, RS485/Modbus, IR protocols, SD card logging, buck-boost regulation, BLDC control, protection algorithms, and hardware bring-up. Delivered production firmware for major brands including **Voltas, Panasonic, and BPL/Reliance**. Promoted to Assistant Manager – R&D for leading architecture design, project execution, and mentoring junior engineers.

TECHNICAL SKILLS

Microcontrollers & Platforms

AVR, STM32, ESP32/ESP8266, TI MSP430/MSPM0, Nuvoton, Holtek

Tools & IDEs

Keil, STM32CubeIDE, Code Composer Studio, Atmel Studio, Proteus, Git, Gitea, Redmine

Languages & Protocols

C, C++, UART, I2C, SPI, RS232, RS485, Modbus, MQTT, HTTP/HTTPS, TCP/IP

Technologies

Wi-Fi, Bluetooth, GPRS/GPS, PWM, IR, Graphic Displays, BLDC Fan Control

EXPERIENCE

Firmware Engineer – R&D

IoTfy Solutions Pvt. Ltd

📅 12/2024 - Present 📍 New Delhi

- Leading R&D for HVAC and smart appliance firmware projects.
- Leading development for flagship client projects: **Voltas VRF ODU, Panasonic AC IDU, BPL/Reliance Washing Machine**.
- Conducting firmware architecture reviews, debugging, testing, and production sign-off.
- Guiding junior engineers and coordinating with hardware, QA and cloud teams.

Embedded Engineer

E4Engineer Pvt. Ltd

📅 01/2023 - 11/2024 📍 Jodhpur

- Developed embedded solutions for industrial monitoring and automation systems
- Delivered end-to-end implementation for IoT-based projects across fuel, telecom, and groundwater sectors
- Configured Gitea and Redmine to streamline development workflow, improving tracking efficiency by 15%

PROJECTS

VRF Outdoor Unit 1OHP(ODU) Control Card

- **Tech Used:** C, RS485, UART, Thermal sensors, BLDC control, EEV stepper control, Protection logic.
- Developed complete outdoor unit firmware enabling ODU-IDU communication, inverter drive interaction, and system coordination.
- Integrated condenser coil, ambient, discharge, and indoor temperature sensing for thermal control.
- Implemented compressor safety protections: high/low pressure, over-temperature, over-current, DC-bus faults, and abnormal power conditions.
- Designed dynamic EEV stepper control and BLDC outdoor fan modulation for optimal cooling efficiency.
- Added RS485 service-tool diagnostics and full VRF operation flow (startup, soft-start, oil-return, steady cooling).

AC Indoor Unit (IDU)

- **Tech Used:** C, IR decoding, PWM, BLDC fan control, UART, Timer modules.
- Developed production-level AC IDU firmware supporting cooling controls, temperature settings, and UI indicators.
- Reverse-engineered and implemented Panasonic long-frame IR protocol with high reliability.
- Integrated BLDC fan control using FG feedback and PWM speed management.
- Added protection logic, swing/flap control, and indoor sensing modules.
- Ensured compatibility with multiple remote commands and mode handling.

Smart Washing Machine

Tech Used: C, ADC, Motor control, Algorithmic load estimation, Water-flow control.

- Implemented sensorless load detection using motor current measurement.
- Automated water fill, agitation, rinse, and spin cycles using algorithmic logic.
- Replaced water-level/weight sensors with embedded logic, reducing BOM cost ~20%.

- Integrated direction control, safety checks, cycle timing, and motor protections.
- Conducted hardware validation and field testing of control sequences.

Air Fryer

Tech Used: C, ADC, Motor control, Algorithmic load estimation, Water-flow control.

- Designed triac-based heater control using temperature-time logic.
- Implemented thermal hysteresis for stable temperature maintenance.
- Integrated user-set temperature and timer management.
- Added safety shutdown for over-temperature and sensor faults.
- Validated heating cycles for consistent performance.

IR Remote System

• **Tech Used:** C, IR TX/RX, NEC protocol, Custom protocol.

- Designed IR remote and receiver hardware.
- Implemented NEC protocol encoding/decoding with precise timing control.
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- Created custom IR protocol for proprietary appliance communication.
- Added noise filtering, low-power handling, and long-range detection.
- Ensured stable 38kHz carrier handling and frame parsing.

Fuel Management System (DG Node)

Tech Used: C, RS485, GPS/GPRS, ADC, IoT communication, Modbus.

- Developed firmware for fuel-level sensing, DG runtime tracking, and generator health monitoring.
- Implemented RS485 communication with Modbus for industrial reliability.
- Integrated GPS/GPRS modules for real-time remote reporting.
- Designed sensor integration for fuel flow, battery voltage, and temperature.
- Built field-ready firmware with auto-reconnect and fail-safe mechanisms.
- Enabled remote diagnostics and reporting for operational efficiency

Underground Water Management System

Tech Used: C, RS485, GPS/GPRS, ADC, IoT communication, Modbus.

- Built IoT monitoring device for water extraction, using flow sensors and pump control.
- Implemented safety shutdown based on extraction limits.
- Logged consumption data and pushed reports to cloud.
- Designed firmware for long-term stable field operation.

Telecom Power & Smart POE System

- **Tech Used:** C, ADC, Buck-Boost Control, Battery Management, DC-DC Regulation, Inverter/Relay Control, Protection Logic.
- Developed firmware for a smart POE/ODU power system capable of generating multiple DC outputs (12V/24V/36V/48V/56V) for telecom routers, ODUs, and POE devices.
- Implemented multi-position knob-based voltage selection, battery backup handling (12V), and automatic AC-fail switchover with stable buck-boost operation.
- Controlled DC-DC modules and inverter logic to maintain uninterrupted power delivery, including AC output generation for routers requiring mains input.
- Added protection and monitoring features including over/under-voltage, overload, short-circuit, reverse-polarity, thermal cutoff, battery voltage sensing, and load current monitoring.

Data Logger

- **Tech Used:** ESP32, SIMCOM7600, SD Card (FAT FS), GPS, Accelerometer, C/C++, AT Commands, Cloud Upload.
- Designed a high-frequency data logging system using ESP32 and SIMCOM7600 for real-time vehicle telemetry and movement tracking
- Implemented SD-card based file rotation logic: device creates 4 CSV files per day, one every 6 hours, named with date-month-year + index (0-3).
- Logged 1-second interval data including timestamp, date, latitude, longitude, altitude, MSL altitude, speed, and accelerometer readings.
- Integrated accelerometer to detect movement, tilt, impact, or sudden motion; all sensor events logged in CSV.

Gitea SelfHost & Redmine Setup

Deployed and configured internal project management and version control tools

- Improved team productivity and code tracking efficiency by ~15%
- Enabled self-hosted version control (Gitea) and full project lifecycle tracking (Redmine)

EDUCATION

Bachelor of Technology (B.Tech) – Mechanical Engineering

Jodhpur Institute of Engineering and Technology, Rajasthan

📅 2019–2023

Senior Secondary (12th Grade)

📅 **BPIC, Bihar**

📅 2019

Secondary (10th Grade)

📅 **D.M. Academy, Bagaha, Bihar**

📅 2017